LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Unit Test: Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations wherever asked. Mention physical states where possible.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following is a correctly balanced chemical equation? **[1]**

(a) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ (b) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

(c) $3Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$ (d) $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Q2. When dilute hydrochloric acid is added to zinc granules, the gas evolved is: **[1]**

(a) oxygen (b) hydrogen

(c) chlorine (d) carbon dioxide

Q3. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ takes place in sunlight. This is an example of: **[1]**

(a) combination reaction (b) thermal decomposition

(c) photolytic decomposition (d) displacement reaction

Q4. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, the substance that gets oxidised is: **[1]**

(a) CuO (b) H_2

(c) Cu (d) H_2O

Q5. Which of the following is an exothermic process? **[1]**

(a) Electrolysis of water

(b) Respiration

(c) Decomposition of calcium carbonate

(d) Photosynthesis

Q6. **Assertion (A):** The blue colour of copper sulphate solution fades when an iron nail is dipped in it. **Reason (R):** Iron is more reactive than copper and displaces copper from copper sulphate solution. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why should a magnesium ribbon be cleaned with sandpaper before burning it in air? Write the balanced equation for its burning. **[2]**

Q8. What is a precipitation reaction? Give one example with a balanced chemical equation. **[2]**

OR

Why are bags of chips flushed with nitrogen gas? Name the process this prevents.

SECTION C

6 marks

Q9. A solution of a substance 'X' is used for whitewashing walls. **[3]**

(i) Name 'X' and write its chemical formula. **[1]**

(ii) Write the balanced equation for the reaction of X with water. **[1]**

(iii) Why does the wall get a shiny finish two to three days after whitewashing? **[1]**

Q10. State the type of each of the following reactions: **[3]**

(i) $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$ **[1]**

(ii) $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$ (on passing electric current) **[1]**

(iii) $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$ **[1]**

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What is a redox reaction? **[1]**

(ii) In the reaction $ZnO + C \rightarrow Zn + CO$, identify the substance oxidised and the substance reduced. Give reasons. **[2]**

(iii) What is corrosion? Give two ways to prevent the rusting of iron. **[2]**

OR

(a) What do you observe when ferrous sulphate crystals are heated in a boiling tube? Write the balanced equation and name the type of reaction. [3] (b) Why is respiration considered an exothermic reaction? Write the equation. [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. In a class activity, students heated lead nitrate powder in a boiling tube. Brown fumes were given off and a yellowish residue was left behind. In another observation, a copper vessel kept in the open for a few months developed a green coating on its surface. **[4]**

(i) Name the brown gas given off when lead nitrate is heated. **[1]**

(ii) What type of reaction is the heating of lead nitrate? **[1]**

(iii) What is the green coating on the copper vessel, and why does it form? **[2]**

~ All the best! ~